

Results

3.1 Cis-regulatory annotation of the feline renal cortex

Combining the chromatin and expression datasets required resolving an assembly discordance that produces no error and no warning. The replicated peak sets of GSE182952 were called against *F.catus_Fca126_mat1.0*, whereas the expression study used *Felis_catus_9.0*. Both assemblies name their chromosomes identically, so translating peak coordinates onto a *Felis_catus_9.0* annotation retained 99.96% of peaks and appeared successful. Permutation testing showed that it was not: promoter H3K4me3 overlap under that harmonisation reached 1.55-fold over the permuted expectation, with H3K27ac at 1.15-fold and CTCF at 1.34-fold, values indistinguishable from the coincidence expected when intervals and annotation occupy unrelated coordinate systems. Annotating on the assembly of the chromatin data instead raised promoter H3K4me3 enrichment to 8.59-fold, with H3K27ac at 3.92-fold and CTCF at 5.09-fold (Figure 1A, Table 2). Because the expression matrices are indexed by gene symbol rather than by coordinate, this substitution carries no cost: gene symbols matched the RefSeq annotation for 99.87% of cortical and 99.88% of medullary genes.

After coordinate translation and removal of the widest 0.5% of intervals, 29,718 H3K4me3, 81,182 H3K27ac and 60,277 CTCF peaks were retained, corresponding to 99.46%, 99.48% and 99.46% of the deposited sets. Promoter marks covered close to half of annotated genes, with H3K4me3 present at the promoters of 17,419 genes and H3K27ac at 17,256.

Delimiting distal regulatory space by CTCF occupancy rather than by a fixed window produced domains with a median width of 39.4 kb and an interquartile range of 12.8 to 102.3 kb (Figure 1C). A minority of genes reached the 250 kb cap imposed to bound domains in CTCF-poor regions, visible as the terminal accumulation of the distribution. Classifying genes by the marks present at their promoter and within their domain assigned 15,586 genes to an active promoter state, 1,591 to a primed promoter state carrying H3K4me3 without acetylation, 10,057 to a state in which the only active mark is a distal enhancer, and 7,513 to no active mark (Figure 1B).

3.2 Differential expression reproduces the source study

The analysed cohort comprises 22 cats, 17 of which contributed both cortex and medulla, with four contributing cortex only and one medulla only (Table 1). Age, sex and breed did not differ across disease groups in either compartment. Age ranged from 2 to 20 years, and three animals were younger than seven, including one cat of two years in the CKD 3/4 group and one of three years in CKD 1/2, a distribution that indicates aetiological heterogeneity within a cohort otherwise characteristic of age-associated disease.

Differential expression computed with limma on the deposited log2 counts per million recovered the compartment asymmetry described in the source study (Figure 2A, Table 1). In the renal cortex, the early contrast yielded 3 genes at a false discovery rate below 0.10 while the late contrast yielded 5,087; in the medulla, the early contrast yielded 1,908 and the late contrast 2,821. The source study, using a quasi-likelihood negative binomial model, reported 6 genes for the early cortical contrast, nearly 2,000 for the early medullary contrast and more than 4,000 for the late cortical contrast. Agreement across a different statistical framework applied to the same normalised matrices indicates that the compartment asymmetry is a property of the data rather than of the analytical engine.

The early cortical contrast is worth noting for a different reason. A previously published secondary analysis of the same accession, which requantified the raw reads with pseudo-alignment and applied a negative binomial model at a stricter significance threshold, reported 539 genes for that contrast. Our value of 3 and the source study's value of 6 agree with each other and differ from that estimate by two orders of magnitude. The comparison is not a clean adjudication, since the three analyses differ in quantification, model and threshold, but it indicates that the early cortical signal is not a stable object.

The contrast between disease stages was near-empty in both compartments, with no gene in the cortex and 8 in the medulla. Because the late cortical contrast is large while the early one is not, and neither differs from the intermediate group, the pattern is consistent with high within-group dispersion at CKD 1/2 rather than with absence of biological difference. This contrast was not carried into subsequent analyses.

Adding age to the model left the cortical analysis essentially unchanged, with a Jaccard index of 0.94 between gene sets and a correlation of 0.999 between fold changes. In the medulla the fold changes were equally stable but the number of genes detected rose from 1,908 to 3,578, with every gene of the unadjusted analysis retained. Age therefore explains an appreciable share of residual variance in the medulla. Because all subsequent analyses use the cortex, and because the source study did not adjust for age, the unadjusted model is reported throughout and medullary counts are conservative.

The volcano distribution of the late cortical contrast (Figure 2B) shows comparable numbers of induced and repressed genes with no asymmetry in effect magnitude, which motivated analysing the two directions separately in all subsequent steps.

3.3 Coexpression structure is not reproducible, and the single stable axis tracks tubular content

Applying hierarchical clustering with dynamic tree cutting to the 1,798 cortical genes passing both a false discovery rate below 0.05 and an absolute log₂ fold change above 1 yielded 13 candidate modules. Under subsampling of animals, only two exceeded a mean Jaccard index of 0.6, and both consisted of repressed genes: a 396-gene module at 0.731 and a 78-gene module at 0.631 (Figure 3A, Table S1). The most stable induced module reached 0.565 and the majority fell between 0.19 and 0.39. Both retained modules were associated with disease group, with first principal components explaining 78.4% and 60.4% of the variance of their member genes.

The asymmetry between directions is not explained by measurement precision. Repressed genes were expressed at higher levels than induced genes (median average expression 4.61 against 3.00), but module stability was unrelated to module-level abundance across the 13 modules (Spearman $\rho = -0.06$, $P = 0.85$). The dissociation is visible in individual cases: the stable 78-gene repressed module had one of the lowest median abundances of the set, while an unstable 90-gene induced module had one of the highest.

The eigengene of the larger stable module correlated with the proximal tubule marker score at $\rho = 0.891$, and inversely with the immune and fibroblast scores at -0.809 and -0.869 (Figure 3B). A module comprising 69% of the repressed programme, with a single component accounting for 78% of its variance and tracking epithelial content this closely, is more parsimoniously described as the proportion of surviving tubular epithelium in the cortical fragment than as a coregulated transcriptional programme. The smaller module was less strongly associated with epithelial content, at $\rho = 0.484$.

Composition and disease stage are nevertheless not collinear at the level of animals: disease group explained 29.0% of the variance in the cortical tubule score, so cats at

comparable stages differed appreciably in tubular preservation. Nor is composition explained by age: none of the three marker scores varied with age, either marginally or conditional on disease group, with point estimates close to zero despite a cohort spanning 2 to 20 years.

Because the medulla shows differential expression at the early stage in a compartment that has not undergone the structural loss dominating the late cortex, the same procedure was applied there. Among the 1,908 genes of the early medullary contrast, 21 candidate modules were detected and none reached the stability threshold, the highest reaching 0.442. Restricting to the 141 genes passing the stricter effect-size filter produced a single module per direction, a configuration in which the subsampling test is uninformative because each module can only be matched against itself. Medullary epithelial content did not differ across disease groups (18.5% of variance explained, $P = 0.215$), confirming that the compartment had not lost structure, but the induced medullary genes tracked the immune marker score at $\rho = 0.831$.

Across both compartments, therefore, the only reproducible structure in the bulk transcriptome corresponds to cellular composition.

3.4 Motif enrichment is largely null, and the ERR observation reduces to promoter architecture

Motif enrichment was tested across two compartments, two directions and two region types, using 401 of 633 JASPAR vertebrate matrices whose corresponding factor was expressed in the compartment analysed, against a background of regulatory regions belonging to expressed but unchanged genes. Of 4,812 tests, seven reached a corrected P below 0.05, all in promoters and none in enhancers, with $\log_2$ enrichment values not exceeding 1.06. This distribution is close to what repeated testing of null hypotheses would produce, and no enhancer analysis yielded a single significant motif.

Within that near-null background, the promoters of repressed cortical genes were enriched for nuclear receptor half-site motifs. Two members of the oestrogen-related receptor family showed enrichment together with reduced expression of the corresponding transcript: ESRRB at 0.87 $\log_2$ enrichment with the factor at $-1.24 \log_2$ fold change, and ESRRB at 0.48 with the factor at -0.62 . Two further nuclear receptor motifs in the same set, NR2F2 and RARA::RXRG, were enriched while their factors increased, and were excluded by the directional coherence criterion. The promoters of induced genes in both compartments were enriched for interferon regulatory and signal transducer motifs, consistent with the inflammatory character of that programme.

Partitioning repressed cortical genes by the presence of an ERR motif in the marked portion of their promoter separated 136 of 2,158 genes. Against a background of all repressed genes, that subset was enriched for respiratory chain complex, mitochondrial inner membrane, aerobic respiration and oxidative phosphorylation, while the same procedure applied to CTCF and SP1 motifs returned no enriched term. The partition was defined by regulatory sequence and the functional annotation was applied afterwards, so the coherence is not a consequence of gene selection.

That result does not, however, establish regulation. Promoters of mitochondrial genes carry ERR elements constitutively, and the association between ERR sites and mitochondrial annotation was present among genes that did not change: the odds ratio was 2.34 among repressed genes and 1.76 among the remainder, with no evidence of heterogeneity between strata ($P = 0.213$), and ERR-positive genes among the unchanged were themselves enriched for aerobic respiration and the citrate cycle. The functional enrichment of the ERR-positive repressed subset is therefore a property of promoter architecture rather than of the disease state.

Inverting the outcome recovers a residual association. Modelling repression as a function of ERR-site presence across 10,959 genes with at least one scanned promoter region, conditioning on mitochondrial annotation and on the amount of promoter sequence scanned, gave an odds ratio of 1.42 for the ERR term ($P = 8.4 \times 10^{-4}$; likelihood ratio $P = 0.0011$), with no interaction with mitochondrial status. Genes carrying an ERR site were also no more strongly repressed than other repressed genes (median log2 fold change -0.69 against -0.75 , $P = 0.16$), so the site is associated with membership in the repressed programme rather than with the magnitude of repression.

3.5 Repressed genes are regulated by kidney-specific enhancers and induced genes by ubiquitous ones

Scoring each kidney regulatory element by the number of the seven profiled tissues in which a corresponding peak was called reproduced the expected contrast between mark classes: 64.4% of H3K4me3 peaks were present in six or seven tissues against 26.8% of H3K27ac peaks, while 20.0% of H3K27ac peaks were confined to the kidney against 10.6% of H3K4me3 peaks (Figure 4A, Table 3). Promoter marks are largely constitutive and enhancer marks largely tissue-restricted, which serves as an internal control on the classification before it is applied.

Genes differed by expression state in the tissue breadth of the enhancers within their CTCF-delimited domains (Figure 4B). Median breadth was 2.82 tissues for repressed genes, 3.80 for unchanged genes and 4.33 for induced genes, with the three distributions differing overall ($P = 1.2 \times 10^{-49}$) and each disease-associated group differing from the unchanged set. In logistic regression adjusted for the number of enhancers per gene, the fraction of a gene's enhancers confined to the kidney was associated with repression at an odds ratio of 2.27 (95% CI 1.86–2.77; $P = 5.2 \times 10^{-16}$) and inversely with induction at 0.33 (0.25–0.43; $P = 3.1 \times 10^{-15}$) (Figure 4C, Table 4). These odds ratios compare the extremes of the range, a gene whose enhancers are all kidney-specific against a gene with none; the median fraction is zero in all three groups, so the effect is carried by the minority of genes that possess such elements.

Two features of the peak data could produce this association artefactually, and neither does. Peak signal intensity correlates with apparent tissue breadth ($\rho = 0.423$), since weakly marked elements are less likely to be detected in other tissues and therefore appear kidney-specific. Enhancer signal intensity was nevertheless almost identical across the three expression states (median fold-enrichment 4.27, 4.18 and 4.21), and adding it as a covariate left the estimates unchanged, at 2.25 for repression and 0.30 for induction. Separately, the kidney has the largest H3K27ac peak set of the seven tissues, exceeding the smallest by a factor of 2.30 (Table S2), so the absolute figure of 20.0% kidney-specific enhancers is an upper bound; this inflation applies uniformly to all kidney peaks and cannot generate a differential association between gene classes.

The direction of the association is consistent across every analysis in this study. Genes that lose expression in the diseased cortex are those whose regulation depends on elements active in the kidney and nowhere else, and genes that gain expression are those regulated by elements shared across organs. Measured in chromatin from healthy animals, without reference to the expression matrix, this is the same phenomenon that the composition analysis describes in expression terms.